Features of OOP

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Features of OOP:

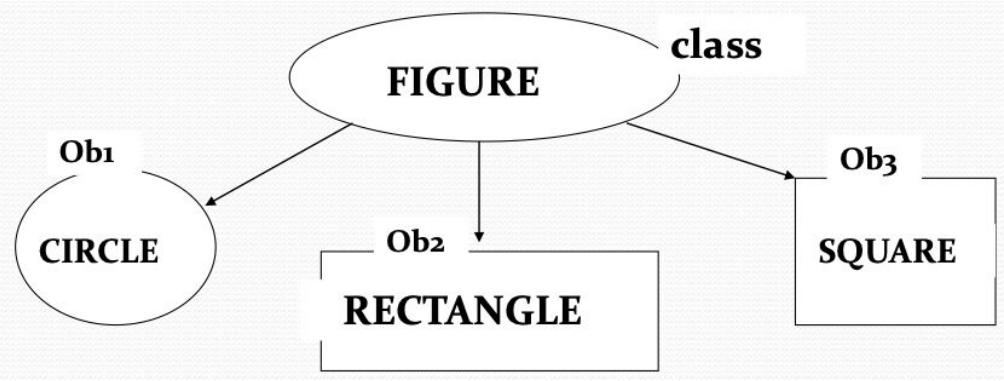
- Object
- Class
- Data Abstraction & Encapsulation
- Inheritance
- Polymorphism
- Dynamic Binding
- Message Passing

Object:

- Objects are the basic run time entities in an object-oriented system.
- They may represent a person, a place, a bank account, a table of data or any item that the program has to handle.

Class:

- Class is a blueprint or an idea of an Object.
- From One class any number of Instances can be created.
- It is an encapsulation of attributes and methods.
- When no access modifier is specified for a class, method, or data member – It is said to be having the **default** access modifier by default (Accessible **only within the same package).**



Data Abstraction:

- It refers to the act of representing essential features without including the background details or explanations.
- For Example: **Car Driving.**
- Achieved using **Abstract** keyword.

Inheritance:

- Inheritance allows to reuse classes by deriving a new class from an existing one.
- The existing class is called the **parent** class, or superclass, or **base** class
- The derived class is called the child class or **subclass**.
- The child class inherits characteristics of the parent class(i.e the child class inherits the methods and data defined for the parent class.
- Achieved using **extends** keyword.

Polymorphism:

- “Poly” means many and “morphism” means form i.e.. many forms.
- Polymorphism means the ability to take more than one form.
- For example, an operation have different behavior in different instances.
- Achieved using Method **Overloading** and Method **Overriding**.
- **Method Overloading:** If the same class implements the same method multiple times.
- **Method Overriding:** If the child class implements the same method present in the parent class.

Dynamic Binding:

- Binding refers to linking of procedure call to the code to be executed in response to the call.
- **Dynamic binding** (or late binding) means the code associated with a given procedure call is not known until the time of call at run time.
- **Static binding** means the code associated with a given procedure call is known at the time of compilation.

Message Passing:

- An object oriented program consists of set of object that communicate with each other.
- Objects communicates with each other by sending and receiving information .
- A message for an object is a request for execution of a procedure and therefore invoke the function that is called for an object and generates result